

Unit Topic: Scientific Method Honors

Essential Question: How does the nature of science, scientific method, dimensional analysis, scientific notation, lab equipment and safety serve as a foundation for what we learn about in physical science throughout the year?

Concept 1: <i>Lab Basics: Safety and Equipment</i> Objectives: <i>Identify safe and unsafe behaviors in a given scenario.</i> <i>Identify and describe the use of lab equipment given a picture or actual piece of equipment.</i> Vocabulary: Beaker Erlenmeyer flask Test tube Graduated cylinder Watch glass Mortar and pestle Funnel Pipette Beaker tongs Spatula Digital scale Bunsen burner Hot plate Practice: - Describe the uses of the following safety equipment: eye wash, chemical/safety shower, fire blanket, and fire extinguisher. - Describe the appropriate way to smell an unknown substance in lab.	Concept 2: <i>Measurement</i> Objectives: <i>Use an example to explain the difference between accuracy and precision.</i> <i>Calculate an average.</i> <i>Given data, be able to determine if it is accurate, precise, or both.</i> <i>Convert temperature measurements into Celsius, Fahrenheit, or Kelvin.</i> <i>Convert metric measurements.</i> <i>Make precise and accurate measurements with various lab equipment.</i> Vocabulary: Accuracy Precision Meniscus Practice: - List the SI units for length, mass, time and temperature. - The following data was collected for a runner's speed: 4 mi/hr, 3 mi/hr, 6.9 mi/hr, and 7 mi/hr. - Is this data precise? Explain. - If the runner tends to run 5 mi/hr, is this data an accurate representation of the runner's ability? Explain. - Convert the following values: 150 cm to mm 1,200 ks to s - Convert the following values: 20°C to °F 300 K to °C 98°F to K	Concept 3: <i>Dimensional Analysis and Scientific Notation</i> Objectives: <i>Use dimensional analysis to convert numbers into different units.</i> <i>Convert numbers in standard notation into scientific notation.</i> <i>Convert numbers in scientific notation into standard notation.</i> Vocabulary: Conversion factors Practice: - Use dimensional analysis to convert the following values. 191 days to seconds 32 lbs to grams 21 mm to yards - Write the following values in scientific notation. 787,000 4,990,000 0.0413 0.00000011 - Write the following values in standard notation. 6.9×10^{-3} 7.008×10^6 1.02×10^2 5.0553×10^{-4}	Concept 4: <i>Scientific Method</i> Objectives: <i>Make qualitative observations, quantitative observations, and inferences about a picture or scenario.</i> <i>Given a sample experiment, be able to write a hypothesis and identify the independent variable, dependent variable, constants, control group, and experimental group(s).</i> <i>Explain the importance of both constants and a control group to an experiment.</i> Vocabulary: Science Observation Inference Law Theory Hypothesis Independent variable Dependent variable Constant Control group Experimental group Quantitative Qualitative Practice: - List the six steps in the scientific method. Beside them list the nine components of a lab report. - Outline a sample experiment you could set up to investigate the effect of drinking Starbucks on student test scores. List all components in the 2nd objective above. - List the type of graph that would be most appropriate in the following scenarios: - Percentage of students in each grade in the school. - The change in mass of firewood before heating and after. - The effect of the amount of conditioner used on hair length.
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Lab Stations: Safety and Equipment

Travel to each station in the room and write your answers to the questions at each station below. You do not have to go in order, just make sure you get to every station!

<u>Station 1</u>	<u>Station 2</u>
<u>Station 3</u>	<u>Station 4</u>
<u>Station 5</u>	<u>Station 6</u>
<u>Station 7</u>	<u>Station 8</u>
<u>Station 9</u>	<u>Station 10</u>
<u>Station 11</u>	<u>Station 12</u>

Concept 1 Notes: Lab Basics – Safety and Equipment

Questions/Vocabulary

Lab Safety

General Guidelines

- Read lab through ALL THE WAY before doing anything else
- Keep lab table or desk neat and organized to avoid spilling or breaking anything while working
- In case of emergency, always tell the teacher before doing anything else!
 - *No matter how minor, I want to know!!*
- Take lab seriously! Never goof around.
- When in doubt, ask the teacher!

Equipment

- Safety Goggles: always wear when using chemicals or fire to protect your eyes
 - These stay on, even through clean up!
- Lab Apron: wear to protect clothing
- Heat-resistant gloves: wear when using hot objects to avoid burning yourself!

Fire

- To prevent fires:
 - Never walk away from an open flame or a plugged in hot plate
 - Keep hair tied back when working in lab
 - Do not wear loose clothing or jewelry in lab
- A fire extinguisher can be used to put out fires
 - PASS = Pull, Aim, Squeeze, Sweep
 - A *fire blanket* is best to use when a person is on fire

Chemicals

- No eating and drinking in lab!!
 - Don't ever taste a chemical
 - When smelling, make sure to waft
- Wear gloves when handling toxic chemicals
- Always dispose of chemicals correctly
 - Ask before pouring down the sink!!
 - *Tell the teacher if you spill!*
- If you get a chemical on yourself...
 - Use eye wash or safety shower

Glassware

- Hot glass looks just like cold glass
 - *Be careful, and always use safety equipment to handle!*
- Broken glass goes in special disposal, not the trash can
 - *Notify me if your glassware is cracked before using!*
- Closed toed shoes and long pants are the safest attire for working in lab

Practice Time: *List the location of the safety equipment in this classroom.*

Lab Equipment

Each piece of lab equipment is designed for a specific job.

Mortar and Pestle: for grinding solids into a powder

Watch glass: can be used as a lid on a beaker to prevent splattering

Glassware

Graduated cylinder: makes precise volume measurements of liquids

Erlenmeyer flask: Heating liquids, but can also measure and hold them

Beaker: Holding, mixing, heating, and measuring liquids (and some solids)

Test tubes: Used for performing small chemical reactions

Test tube rack: Holds up multiple test tubes at one time

Test tube holder: to hold test tube when heating

- *Always angle the open end of test tube away from you and others while heating*

Heating

Hot plate: electricity powered plate for heating chemicals

Bunsen Burner: gas powered flame for heating chemicals

Ring Stand: Used to support heated objects above the flame

Iron ring: Used to support heated objects above the flame

Wire gauze: Used to support heated objects above the flame

Measuring

Digital scale: Measures mass (usually in grams)

Meter stick: Measures length in mm, cm, or m

Thermometer: Measures temperature in degrees Fahrenheit or Celsius

Transferring

Beaker tongs: moving and holding a hot beaker

Funnel: transferring liquids from larger containers to smaller

Spatula: transferring solid chemicals from larger containers to smaller

Pipette: transferring liquids in small precise amounts

Dropper: transferring liquids drop by drop

Summary Concept 1

Practice: Lab Safety Scenarios

Read each scenario below. Identify what is inappropriate lab behavior in each instance. Then write the appropriate lab safety technique. You may use the back of this page or another piece of paper, if needed.

1. It is lab day in science class! Mr. Benson is out in the hall talking with other teachers when the bell rings. Justin decides to go ahead and get started with the lab. He begins weighing chemicals, touching them with his hands. He gets an itch in his nose and rubs it with his hand.
2. Jane is doing a water displacement lab and drops a metal rod into a graduated cylinder with water, cracking the bottom of the glass cylinder. The base of the graduated cylinder is plastic though, so it is preventing anything from leaking out, so she decides to just keep using it for the rest of the class.
3. Kendra is heating a test tube for an experiment using a Bunsen burner. She didn't put on safety goggles since she already wears glasses. She tilts the test tube mouth away from her towards her lab partner, since the steam keeps fogging up her glasses.
4. Mason got to class late because he had a dentist appointment. He missed reading the lab and going over it, but he decides to jump in with his lab group any way so he doesn't get further behind, since he knows they can catch him up.
5. Eden woke up early to curl her hair before school, and decides to leave it down during her chemistry lab because she doesn't want to get a crease in it from her hair tie. She volunteers to help with the Bunsen burner, leans forward, and catches the end of one of her long curls on fire. Her lab partner Theo volunteers to get the fire extinguisher to spray her down.
6. Thomas and Trenton aren't finished with their lab when the bell rings, but they don't want to be late to lunch. They quickly pour their leftover chemicals down the drain and leave the rest of the lab equipment they used on the counter, figuring someone in the next class period will probably be using it anyway.
7. Lexie's teacher keeps going on and on about lab safety, and where all of the safety equipment is in the classroom, but she keeps daydreaming about going home to watch Netflix all afternoon when school gets out. She figures it doesn't matter anyway because she'll never need to use any of the safety equipment. It's not like anything bad will ever happen to her.
8. JD is using beaker tongs to move a beaker around on a hot plate to heat the contents within the beaker. He's afraid to break the beaker, so he uses the beaker tongs to hold the beaker at the very bottom. Each time he moves the beaker around, the rubber part of the beaker tongs rubs along the surface of the hot plate. The room starts to smell very funny, but no one thinks anything of it.

Practice: Lab Equipment Identification

Part 1: List which piece of lab equipment would be most useful for each of the following tasks.

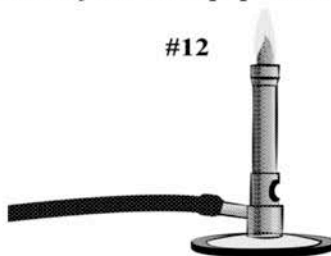
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|---|---|
| 1. Hold 50 mL of boiling water. | 6. Keep contents in a beaker from splattering when boiling chemicals. |
| 2. Crush a chunk of chemical into a powder. | 7. Find the temperature of an unidentified substance. |
| 3. Heat a beaker of water when a gas line is not available to power any equipment. | 8. Transfer 3 drops of indicator into a solution. |
| 4. Measure exactly 42.5 mL of water. | 9. Hold 4 different test tubes needed for an experiment. |
| 5. Scoop ammonium nitrate out of its container and onto weigh paper to be measured. | 10. Find the mass of 125 grams of sodium chloride. |

Part 2: Label each picture with the correct name of the lab equipment.

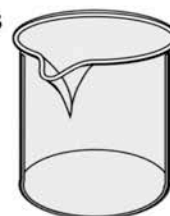
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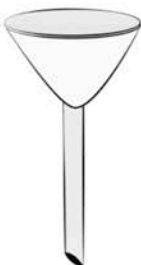
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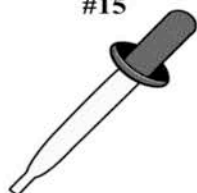
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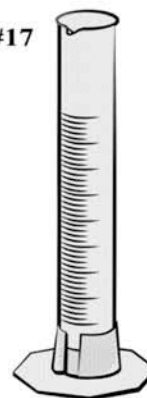
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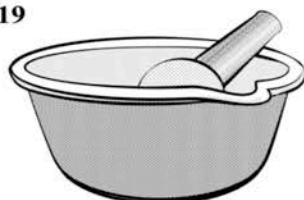
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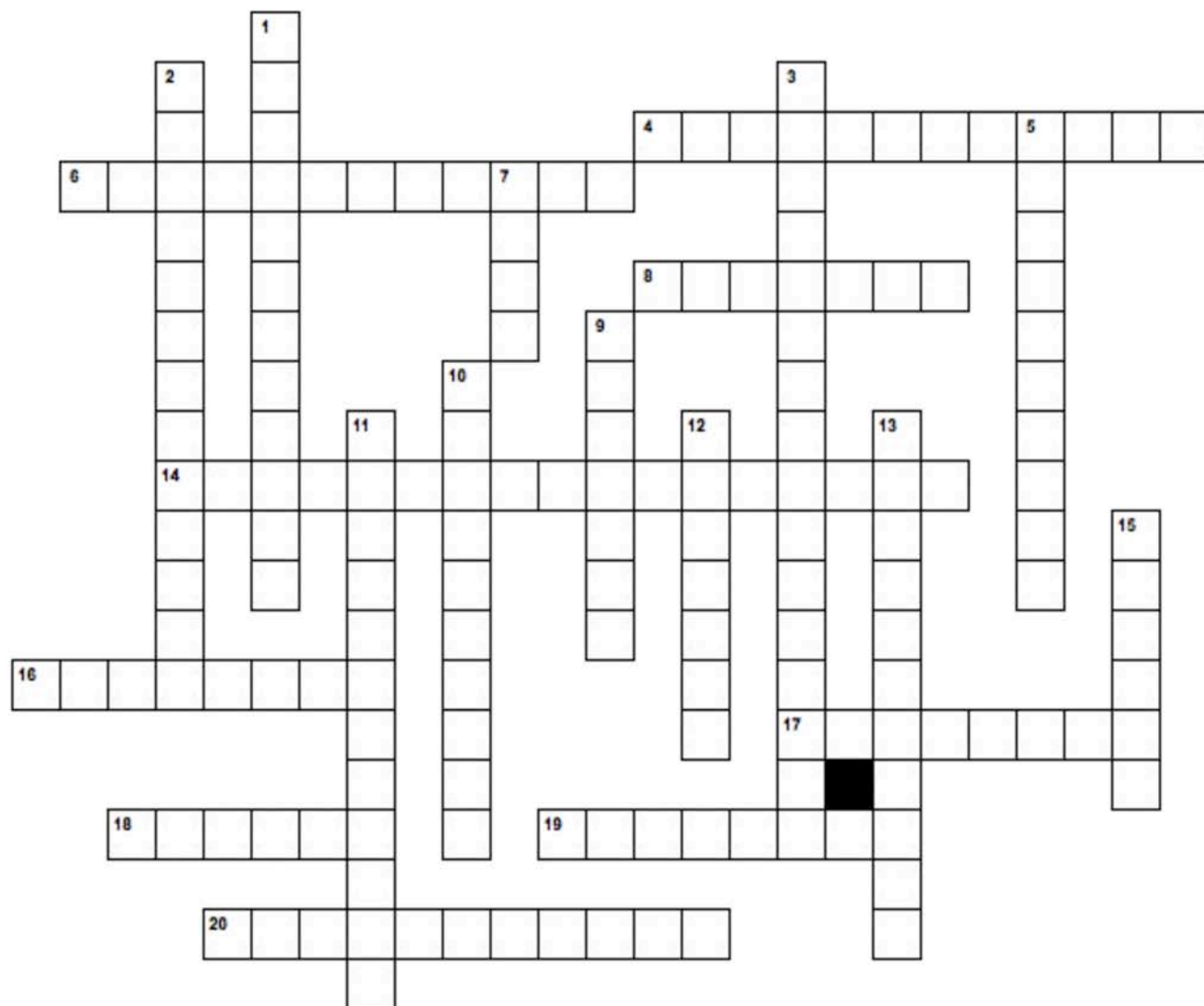
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Activity: Lab Safety and Equipment Puzzle



ACROSS:

4. Used to measure mass
6. Used only if you spill chemicals on your body
8. Used to transfer precise amounts of liquids
14. Used to measure precise volumes of liquids
16. Where you should NEVER dispose of broken glass
17. Used to heat chemicals; powered by electricity
18. Used to heat, mix, hold, and measure chemicals
19. Worn to protect clothing in lab
20. Best for putting out a fire on a person

DOWN:

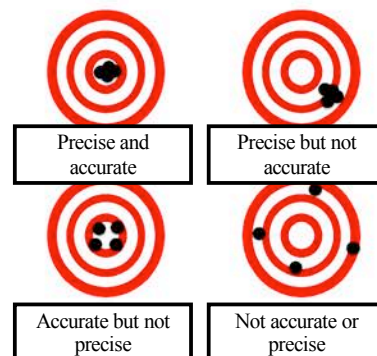
1. Used to hold test tubes in an upright position
2. Worn to protect eyes from fire and chemicals
3. To operate you must Pull, Aim, Squeeze, Sweep
5. Type of shoes to be worn in lab
7. Appropriate way to smell chemicals in lab
9. When in doubt, ask this person
10. Used to measure length
11. Used to heat chemicals; powered by gas
12. What you should always do at the end of lab
13. Used to move glassware that's been heated
15. Used to transfer liquids into small containers

Concept 2 Notes: Measurement

Measurement

Accuracy: the degree to which the experimental result agrees with the accepted value

- How correct the measurement is



Precision: the degree to which measurements agree with one other

- How specific the measurement is; or how consistent multiple measurements are to each other

Example #1: *If a student knows 85% of the material for the test but makes a 70% on the test, is the measurement accurate, precise, or neither?*

- Answer: The measurement isn't precise or accurate

Example #2: *Mrs. Joyner's dog, Millie, is known to weigh 65 lbs. Millie goes to the vet and is weighed three times at 71 lbs, 70.5 lbs, and 71.1 lbs. Are these measurements accurate, precise, or neither?*

- Answer: The measurements are precise but not accurate

Average: add up all values and divide by the total number of values

Example #3: *Callie measures her flower to be 30 cm tall. Jeff measures the plant to be 45 cm tall. Liam measures the plant to be 36 cm tall. Find the plant's average height.*

$$\frac{30 \text{ cm} + 45 \text{ cm} + 36 \text{ cm}}{3} = \frac{111 \text{ cm}}{3} = \boxed{37 \text{ cm}}$$

Standard Units

Standard: an exact quantity that people use for comparison

SI System: the system used by scientists around the world to make measurements

Units

Length: distance between two points

- Unit = meter = m
- *Often measured using a meter stick or ruler*

Mass: amount of matter in an object

- Unit = kilogram = kg
- *Often measured using a digital scale or a spring scale*

Volume: amount of space occupied by an object

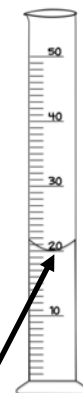
- Unit = Liter = L
- *Often measured using a graduated cylinder or can be calculated with a geometric equation*

Time: interval between two events

- Unit = seconds = s
- *Often measured using a stopwatch*

Temperature: the amount of heat in an object

- Unit = Kelvin = K
 - The metric system uses Celsius ($^{\circ}\text{C}$)
 - The U.S. uses Fahrenheit ($^{\circ}\text{F}$)
- *Often measured using a thermometer*



When measuring with a graduated cylinder, always measure to the bottom of the meniscus (the curve in the liquid)

Temperature Conversions

$$\text{K} = ^{\circ}\text{C} + 273$$

$$^{\circ}\text{C} = 5/9 (^{\circ}\text{F} - 32)$$

$$^{\circ}\text{F} = (^{\circ}\text{C} * 9/5) + 32$$

Example #4: Mrs. Smith is going on vacation to Greece for her anniversary and the weather is predicted to be 27°C . Mrs. Smith lives in the U.S. though and doesn't know what to pack for this weather. Help her determine the temperature in $^{\circ}\text{F}$ so she can pack.

$$^{\circ}\text{C} = 27$$

$$^{\circ}\text{F} = ?$$

$$^{\circ}\text{F} = (^{\circ}\text{C} * 9/5) + 32$$

$$^{\circ}\text{F} = (27 * 9/5) + 32$$

$$^{\circ}\text{F} = (48.6) + 32$$

$$^{\circ}\text{F} = 80.6^{\circ}\text{F}$$

Practice Time!

1. Convert 23°C to $^{\circ}\text{F}$.
2. Convert 48°F to $^{\circ}\text{C}$.
3. Convert 83°F to Kelvin.

Prefixes

These prefixes are used to avoid very large or very small numbers. They attach to the unit name. Ex. Milligram = mg

1 Mega (M) = 1,000,000 units

1 Kilo (k) = 1,000 units

1 hecta (h) = 100 units

1 deka (da) = 10 units

UNIT (m, g, L, s)

10 deci (d) = 1 unit

100 centi (c) = 1 unit

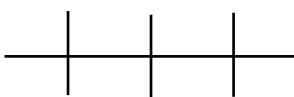
1,000 milli (m) = 1 unit

1,000,000 micro (u) = 1 unit

1,000,000,000 nano (n) = 1 unit

Remember, the unit is either m, L, s, or g depending on what you are measuring

Metric Conversions

1. Write down given number and unit.
2. Draw picket fence. 
3. Use chart to fill in appropriate conversion factors, making sure matching units are on opposite sides of fence to cancel out.
4. Multiply the top line across and the bottom line across.
5. Divide the top by the bottom.

Example #5: How many meters are there in 48 km?

$$\begin{array}{c|c|c} 48 \text{ km} & 1,000 \text{ m} & \\ \hline & 1 \text{ km} & \end{array} = \frac{48,000 \text{ m}}{1} = \boxed{48,000 \text{ m}}$$

Example #6: How many kilograms are there in 45,456 mg?

$$\begin{array}{c|c|c|c} 45,456 \text{ mg} & 1 \text{ g} & 1 \text{ kg} & \\ \hline & 1000 \text{ mg} & 1000 \text{ g} & \end{array} = \frac{45,456 \text{ kg}}{1,000,000} = \boxed{0.045456 \text{ kg}}$$

Practice Time!

1. Convert 6.1 ks to s.
2. Convert 0.859 mg to cg.
3. Convert 65.4 m to mm.

Summary Concept 2

Practice: Temperature Conversions

Complete the following temperature conversions. Make sure to show all work to receive credit! You may need to use another sheet of paper.

1. Convert -21°C into Kelvin.
2. Convert 589 Kelvin into $^{\circ}\text{C}$.
3. Convert 45°F into $^{\circ}\text{C}$.
4. Convert 89°F into $^{\circ}\text{C}$.
5. Convert 0°C into $^{\circ}\text{F}$.
6. Convert 100°C into $^{\circ}\text{F}$.
7. Convert 303 K to $^{\circ}\text{F}$.
8. Convert 58°F to K.
9. Emma wants to go swimming the first day the pool opens in May, but she is worried it will be too cold since summer only just started. She only likes to swim in the water if the pool is at least 82°F . The thermometer in the pool reads 24°C . Is the pool warm enough for Emma to swim in? Show all work to support your answer.
10. Stephen is worried he has a fever. His dad is a scientist though, and only has a thermometer that measures in Kelvin, and it measures Stephen's temperature to be 312 Kelvin. If Stephen's temperature is above 100°F , he isn't allowed to go to school. Can Stephen go to school today? Show all work to support your answer.

Practice: Metric Conversions

Part 1: Fill in the SI unit for each of the following measurements.

1. Time: _____ 2. Length: _____ 3. Mass: _____ 4. Temperature: _____

Part 2: Write in the correct abbreviation for each metric unit below.

5. Kilogram: _____ 6. Milliliter: _____ 7. Gram: _____ 8. Centigram: _____
9. Millisecond: _____ 10. Kilometer: _____ 11. Liter: _____ 12. Decimeter: _____

Part 3: Complete the following metric conversions. Make sure to show all work to receive credit! You may need to use another sheet of paper.

13. Convert 2,000 mg to g.

14. Convert 5 L to mL.

15. Convert 104 km to m.

16. Convert 16 cm to mm.

17. Convert 5.6 kg to mg.

18. Convert 1,000 s to hs.

19. Convert 250 m to km.

20. Convert 1,000 cL to daL.

Lab Stations: Measurement

You can go through the following stations in any order you'd like. Each station is to help you practice using different measuring tools we will be using in this class. Take your time and attempt to make **accurate** and **precise** measurements.

After you make all your measurements, complete the conversions in each box. Make sure to show all work on a separate sheet of paper to receive credit.

Some hints:

- Graduated cylinders should be placed on a flat, level surface and read at eye level from the meniscus (bottom of the curve.)
- Pipettes and droppers have mL markings; be sure to note how many mL the one provided measures before using it.
- String/yarn is provided to help you measure irregularly shaped objects.
- Make sure to always remove the thermometer from the liquid you're measuring after you are done.
- Remember you can zero the scale before measuring or after adding a weigh boat.

<u>Station 1</u> Convert to cL:	<u>Station 2</u> Convert to cm:
<u>Station 3</u> Convert to Kelvin:	<u>Station 4</u> Convert to kg:
<u>Station 5</u> Convert to L:	<u>Station 6</u> Convert to hL:
<u>Station 7</u> Convert to m:	<u>Station 8</u> Convert to mm:
<u>Station 9</u> Convert to mg:	<u>Station 10</u> *Challenge* Convert to hours:

Concept 3 Notes: Dimensional Analysis and Scientific Notation

Questions/Vocabulary

Dimensional Analysis

Purpose: A technique of converting numbers into different units, without changing their value

- Useful when:
 - Baking or cooking
 - Traveling to foreign countries that use different units than us
 - Building or engineering
- Done by multiplying given numbers by conversion factors in order to get number into desired units

Conversion factors: ratios of equivalent values (meaning they equal 1)

- Used in dimensional analysis to alter the number without changing its value

Example #1: How many hours are in a day?

24 hours = 1 day, or...

$\frac{24 \text{ hours}}{1 \text{ day}}$ or $\frac{1 \text{ day}}{24 \text{ hours}}$

Example #2: How many inches in 1 foot?

12 inches = 1 foot, or...

$\frac{12 \text{ inches}}{1 \text{ foot}}$ or $\frac{1 \text{ foot}}{12 \text{ inches}}$

Both of these formats are useful when converting units with conversion factors

Steps to apply this method in a real problem:

1. Write down the given (starting) number and unit.
2. Draw a "picket fence."
3. Use chart to fill in appropriate conversion factors, making sure matching units are on opposite sides of fence to cancel out.
4. Multiply the top line across and the bottom line across.
5. Divide the top by the bottom.

Example #3: How many seconds are in 1 year?

$$\begin{array}{c|c|c|c|c} 1 \text{ year} & 365 \text{ days} & 24 \text{ hours} & 60 \text{ min} & 60 \text{ s} \\ \hline & 1 \text{ year} & 1 \text{ day} & 1 \text{ hour} & 1 \text{ min} \end{array} = \frac{31,536,000 \text{ s}}{1} = \boxed{31,536,000 \text{ s}}$$

Example #4: How many feet are in 250.4 cm?

$$\frac{250.4 \text{ cm}}{2.54 \text{ cm}} \times \frac{1 \text{ inch}}{1 \text{ inch}} \times \frac{1 \text{ ft}}{12 \text{ inches}} = \frac{250.4 \text{ ft}}{30.48} = \boxed{8.22 \text{ ft}}$$

Useful and common conversion factors:

<u>MASS</u>	<u>VOLUME</u>	<u>LENGTH</u>
1000 g = 1 kg 1 ounce = 28.35 g 1 kg = 2.2 lbs 16 ounces (oz) = 1 lb 2000 lbs = 1 ton	1 tsp = 5 mL 1 cup = 236 mL 1 fluid oz = 29.6 mL 1000 mL = 1 L 3 tsp = 1 Tbsp 16 Tbsp = 1 cup 8 oz = 1 cup 2 cups = 1 pint 2 pints = 1 quart 4 quarts = 1 gallon	1 inch = 2.54 cm 12 inches = 1 ft 1 mile = 5,280 ft 3 ft = 1 yard

Practice Time!

1. 3.6 L = _____ gallons

2. 600 g = _____ lbs

3. 100 cm = _____ yds

Scientific Notation

Purpose: A technique used to rewrite very large or very small numbers into a format that is easier to use

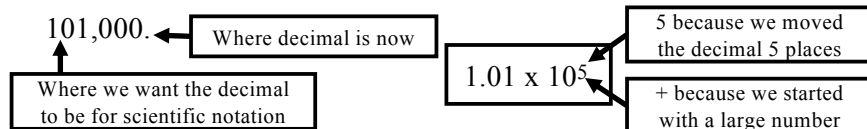
Format: Just the digits with a decimal point after the *first digit*, followed by x 10 to the power which represents how many places the decimal was moved

- Example: 50,500 = 5.05 x 10⁴

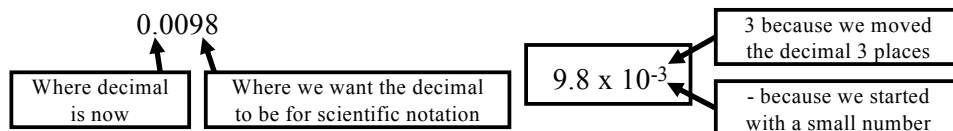
Steps to apply this method in a real problem:

1. Move decimal so that there is only 1 digit in front (to the left) of it.
2. Rewrite the number as #._____ with a **x 10** after.
3. Add an exponent to the 10 to represent the number of places you moved the decimal.
4. Make the exponent:
 - + if you started with a big number (> 1)
 - or
 - if you started with a small number (< 1)

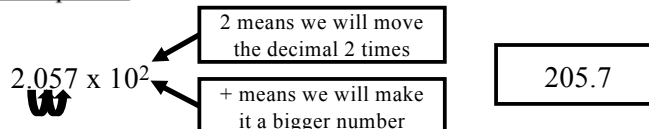
Example #5: Convert 101,000 into scientific notation.



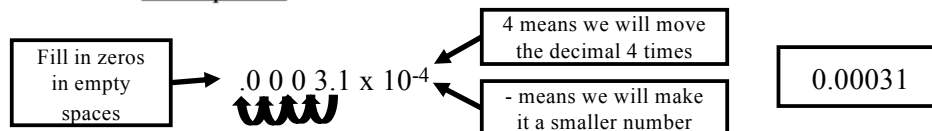
Example #6: Convert 0.0098 into scientific notation.



Example #7: Convert 2.057×10^2 into standard notation.



Example #8: Convert 3.1×10^{-4} into standard notation.



Practice Time!

Place in scientific notation.

Place in standard notation.

1. 354,000,000 = _____ 3. 2.76×10^{-3} = _____

2. 0.000096 = _____ 4. 4.011×10^4 = _____

Explain the purposes for using dimensional analysis and scientific notation, both in the classroom but also in the real world.

Summary Concept 3

Practice: Dimensional Analysis

Show all work as you complete each problem below. Make sure to circle or put a box around your final answer and double check your units!

1. 315 cups to liters
2. 125 lbs to grams
3. 11,900 cm to yards
4. 84 cups to gallons
5. 7 feet to kilometers
6. Gabi has plans with her friends to go to a concert on her birthday in 4 days. She is so excited that she wants to know how many seconds she has left to wait. Can you help her to determine this?
7. Your Uncle Danny just moved into a new apartment and wants to make his famous spaghetti sauce for his new neighbors. Unfortunately, he hasn't unpacked all of his kitchen utensils. His recipe calls for $\frac{1}{2}$ a cup of oregano, however he can only find a teaspoon to measure out his spices. How many teaspoons of oregano would be the same as $\frac{1}{2}$ a cup?
8. Lamar has been running sprints to prepare for his next football game. He has found that he can maintain his maximum speed for 45 yards. He's thinking of running in a 5 km race in a few months, but doesn't know if he can maintain his maximum speed for the entire 5 km. Can you help him determine how far he can?

Practice: Scientific Notation

Part 1: Write the following numbers in scientific notation. Make sure your final answer for each is clearly marked.

1. 0.000000199

2. 24,070

3. 590

4. 0.00308

5. 0.000500

Part 2: Write the following numbers in standard notation. Make sure your final answer for each is clearly marked.

6. 2.105×10^6

7. 1.99×10^4

8. 7.006×10^{-3}

9. 4.11×10^{-5}

10. 5.03×10^2

Lab: Let's Make Fudge

Applying dimensional analysis and scientific notation to the real world

Everyone loves grandma's fudge. The problem is, she always writes her recipes in units that are hard to use in the kitchen when you actually bake! Your job is to convert the measurements below into measurements that you can use to bake a single serving of fudge. First you will use your scientific notation skills to convert the measurements into standard notation. Then use your dimensional analysis skills, and the conversion factors on the next page, to get the measurements into the required units. Round each number to the nearest whole number. Place your final answers in the appropriate box. You must get signed off by teacher before you can make any fudge. Be sure to show ALL work!

Ingredients:

1. 1.71×10^{-1} teaspoons of vanilla into **drops**

Drops of Vanilla

2. 1.32×10^{-3} gallons of milk into **teaspoons**

Tsp of Milk

3. $1.50 \times 10^{-5} \text{ m}^3$ of butter into **tablespoons**

Tbsp of Butter

4. 6.875×10^0 grams of cocoa into **teaspoons**

Tsp of Cocoa

5. 2.75×10^{-2} kilograms of powdered sugar into **tablespoons**

Tbsp of Powdered Sugar

Teacher Signature of Approval: _____

Conversion Factors:

1 gallon = 4 quarts
1 quart = 0.946 L
1 L = 1,000 mL
1 tbsp = 3 tsp
1 tsp = 5 mL
1 mL = 7 drops
1 mL = 1 cm³
1 m³ = 1,000,000 cm³
1 cup = 48 tsp
1 cup = 110 g
1 kg = 1,000 g
1 cup = 200 mL

Procedures:

1. Do nothing until you've received official teacher approval on previous page!
2. Once you have teacher signature, carefully and precisely measure out each ingredient at the measurement station and put into the cup/beaker/can that you are using to make your fudge.
3. Set the hot plate to medium heat.
4. Place your cup/beaker/can onto the hot plate and stir slowly with spoon as it heats.

Special Notes:

- If you are using a plastic spoon, make sure NOT to press it into the cup because it could melt!
 - Make sure to hold the handle of your cup with an oven mitt, or use beaker tongs for cans/beakers because they WILL get hot as they heat!
5. Heat the fudge until it is the consistency/thickness of icing. If you over heat, it will get too syrupy.
 6. Remove fudge immediately from hot plate when it reaches icing consistency. Carefully walk cup of fudge over to the digital scale.
 7. Create a weigh boat out of aluminum foil by folding the edges up so that no hot fudge could pour out. Record the mass of the weigh boat in the data table below.
 8. Pour fudge into weigh boat and let cool.
 9. While fudge is cooling, clean spoon and cup/beaker/can and return it to measurement station.
 10. Once fudge has cooled, record the mass of the fudge.

Special Notes:

- Get the mass of the weigh boat with fudge in it. This is the total mass. Then subtract the mass of the weigh boat from the data table in order to determine the mass of just the fudge. Record in data table.

Results:

Data Table:

	Weight Boat	Fudge	Total (Weigh Boat + Fudge)
Mass (g)			

Analysis:

1. Reflect on what you just did. What did you and your partner do well when making your fudge? What did you not do well?
2. If you wanted to help next year's students do this lab better, what advice would you give them?
3. How are the skills we learned in class of dimensional analysis and scientific notation useful in the real world, based on what we did in this lab? List two additional real world applications for this knowledge.
4. **Think about it:** This lab makes a single serving of fudge. How much of each ingredient would you need if you wanted to use this recipe to make fudge for 5 people instead of just 1 like we did?
5. **Challenge:** Let's say that you decide to make some extra money by selling the fudge you and your classmates made today. If fudge tends to cost about 25 cents per gram, and you make the same amount of fudge every day for the next 30 days, how much money could you make from your fudge this month?

Concept 4 Notes: Scientific Method

Science: a method for studying the natural world by gaining knowledge through observations and investigations

- *Scientists gather information through observations and design investigations based on inferences*

Observation: objective; based on the 5 senses, not opinion

- Ex. The ground is wet.

Inference: subjective; a guess about an observation

- Ex. It rained. Someone spilled something.

Law: description of how things happen in nature

- Ex. Newton's Laws of Motion

Theory: an explanation, based on evidence, of why things happen

- Ex. Evolutionary Theory

- *Laws cannot become theories, and vice versa.*

- *All theories start as hypotheses until substantiated by evidence.*

Scientific Method: technique for investigating phenomenon.

- *Summarized (usually) into 6 steps.*

1. Ask a question.

- Observations we make should lead us to ask questions.
- This clearly stated question will define our purpose for conducting an investigation.

2. Research and gather background information.

- Gather information to see what others have studied about this question, summarizing in an introduction.
 - *What would be trustworthy sources we could look to for useful background information?*
 - *What would be not trustworthy sources to find useful background information?*

3. Form a hypothesis.

Hypothesis: a testable prediction that attempts to answer a question

- Usually predicts a cause and effect relationship between variables

Format: If...(IV), then...(DV), because...

Independent variable: what is being purposefully changed; what is being tested; the “cause”

Dependent variable: what is being measured; the data collected as a response; the “effect”

Example #1: *I always notice that many runners drink Gatorade. I want to know if Gatorade makes them run faster.*

IV: the runner’s drink, Gatorade or water

DV: the runner’s time

Hypothesis: If the runners drink Gatorade, then they will run faster because the electrolytes will give them more energy.

4. Test the hypothesis.

- Design an experiment. List all materials you will need and write out each step in your experiment in a procedure.

A good experiment should always have:

Control group: the normal group used for comparison

- *Ex. runners who drink water*

Experimental group: the group that is being tested and purposefully changed

- *Ex. runners who drink Gatorade*

Constants: factors that do not change in the experiment; conditions kept the same for all groups.

- *Ex. same age, same gender, same training, same shoes/equipment, same track/environment, same length of race, same amount of drink, etc.*

5. Analyze the data.
 - Organize results into tables and graphs

Quantitative: numerical data

Qualitative: descriptive data

3 Main Types of Graphs:

Line Graph: quantitative data vs. qualitative data

- *Ex. Plant growth over time*

Bar Graph: qualitative data vs. quantitative data

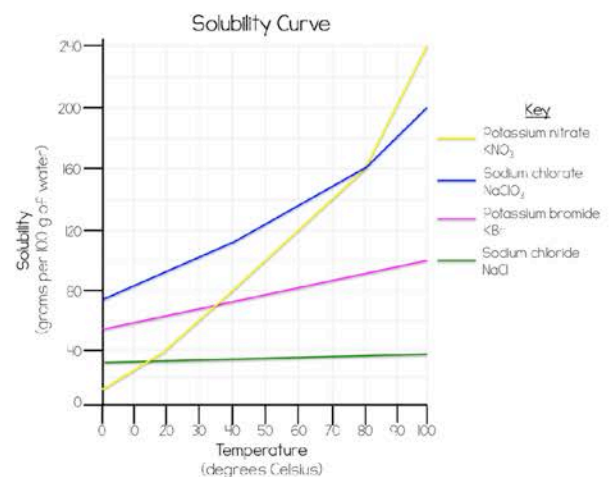
- *Ex. # of students and favorite classes*

Circle (Pie) Graph: percentages

- *Ex. % of students who got A's, B's, C's, D's, and F's on a test*

Graphs Must Have:

- Title
- X-Axis and Y-Axis labels
 - Including units, if needed
- Even scale
- Key, if needed



In a written analysis include:

- A brief overview of the variables
 - *Independent variable, dependent variable, control group, experimental group, and constants*
- Clear explanation of what the data shows
 - *Refer to graphs!*
- Infer why you got the results you did
- Error analysis
 - *Describe at least 3 factors not held constant that could have messed with results and how to prevent in future*

6. Draw conclusions.

- Conclusion should include:
 - A clear statement of if the data supports the hypothesis or not
 - *Never say “proven”, “disproven”, “correct”, or “incorrect”!!*
 - What was learned
 - Real world application

Summary Concept 4

List the steps of the scientific method. Beside them, write how those steps are written out in a formal lab report.

Activity: Candy Graphing

Question: Are bags of candy packaged randomly, or by a machine which ensures the same distribution of candies in each bag every time?

Purpose: To determine if the individual candies in a bag of candy are evenly distributed or not.

Hypothesis:

Materials:

- Regular sized bag of multi-colored candy
- Graph paper
- Ruler
- Colored pencils
- Compass (optional)

Procedures:

1. Pour out candy onto desk. Separate individual candies by color.
2. Count number of candies of each color and record in data table.
3. Calculate the percent of the whole bag represented by each color and record in data table.

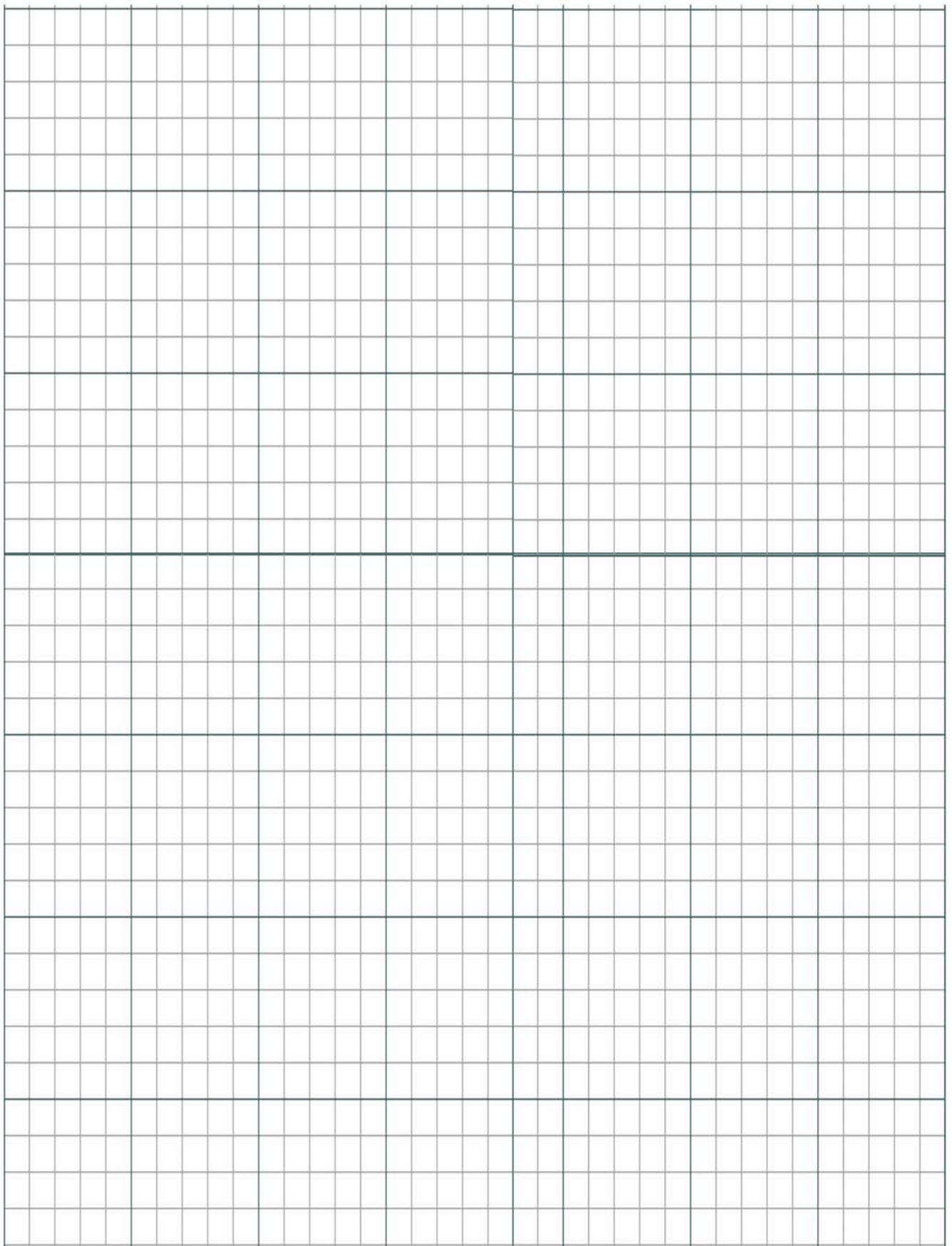
$$\text{Percent} = \frac{\text{Part}}{\text{Whole}} \times 100$$

Data Table:

Color	Number of Candies	Percent of Whole Bag

Results:

- Use the graph paper on the next page to make two graphs – a bar graph and a pie graph.
- **For the bar graph:** Use the first two columns in your data table. Be sure to include all required components of a bar graph (Title, key, labeled axis, etc.)
- **For the pie graph:** Use the percentages in the last column of your data table. Be sure to include a title and to label the percentages on the graph.



Analysis Questions:

1. If the candies were packaged mechanically, what would you expect to see in the distribution?

2. If the candies were packaged randomly, what would you expect to see in the distribution?

3. What did your data show you? What does it suggest?

4. Find someone else in the class who had the same candy as you. Compare your results. Does your data agree? Explain.

Conclusion:

5. Does the data cause you to support or reject your hypothesis? Explain.

Practice: Scientific Method

Read each scenario and answer the questions to the right. Below each description, also label the type of graph that would be most appropriate to represent the results.

Maggie loves fresh brownies, and her grandma told her if she places a piece of bread in the storage container with the brownies, the brownies will stay fresh and moist longer. Maggie decides to test this idea. She bakes a batch of Betty Crocker brownies. She places 6 brownies in 6 individual Tupperware containers. Each brownie is the same size and placed in the same location. She adds a piece of bread to three of the brownie containers, and leaves the other three alone. Each day for 3 weeks she measures the moistness of the brownies, and records her results.	1. Independent variable: 2. Dependent variable: 3. Hypothesis: 4. Control group: 5. Experimental group: 6. Constants:
Alex wants to grow some flowers for his wife, but isn't sure which flowers will grow best in the soil in their yard. He goes to Lowe's and buys hydrangeas, roses, daisies, and tulips. He sections off the soil into four equal parts and plants each type of flower in their own section. All flowers receive the same amount of sunlight throughout the day and he waters them all the same amount and at the same time daily. He records the growth of the flowers once a week for a month.	1. Independent variable: 2. Dependent variable: 3. Hypothesis: 4. Control group: 5. Experimental group: 6. Constants:
Abbey always gets zits on her chin and she can't figure out the best method to get rid of them. She decides to test several products to see which clears her face up fastest: washing with soap and water, washing with an acne face wash from the pharmacy, and washing her face with organic oils. For one week she just washes her face with water. The next week she washes every day with soap and water. The following week she uses the acne wash, and the last week she uses the organic oils. Each week she records the number of zits she has on her chin at the end of the day.	1. Independent variable: 2. Dependent variable: 3. Hypothesis: 4. Control group: 5. Experimental group: 6. Constants: 7. Possible threats to accuracy of results?
Jenny loves to chew gum, but hates losing the flavor. She normally chews Juicy Fruit, but she wants to know if there is another type of gum that holds the flavor longer. She buys 4 different packs of gum, one of them being her normal Juicy Fruit. Every day after lunch she picks one piece of gum and chews it until it loses flavor. She times how long it takes for the gum to lose flavor, and records the time. She chews the same type of gum every day until she has used up the pack, and finds an average time for how long the flavor lasted. Then she starts chewing a new type of gum, and continues until all 4 types have been tested.	1. Independent variable: 2. Dependent variable: 3. Hypothesis: 4. Control group: 5. Experimental group: 6. Constants: 7. Possible threats to accuracy of results?